

Continuous Distillation

Pre-Lab Plan
Unit Operations Lab 3
02/18/2020

Commented [PTH1]: 29/35

Group 2, Tuesday, PM section

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1. Experimental Objective:

The purpose of this experiment is to study the behavior of separation of binary mixture of water and ethanol using binary distillation column. An important process variable in this experiment is the reflux ratio at the top of the column. Furthermore, it is essential role in controlling the product purity and in determining the energy which is required in the reboiler. Understanding how column plate efficiency and the optimal number of stages is the goal for this experiment. Thus, the collected data will be graphed on McCabe-Thiele curve for Binary Distillation to explore the effects of the changing process variables.

In addition, understanding how continuous distillation columns operate gives incite on how chemicals like ethanol, common gas, diesel, and jet fuel are used to separate mixed feed streams into their own distinct products through a continuous distillation.

2. Experimental:

2.1 Apparatus:

The Armfield UOP3CC Continuous Distillation Column is a self-contained distillation facility consisting of two interconnected units; a floor-standing process unit and a bench-mounted control console. The distillation column apparatus or process unit from Figure 1 is connected and controlled by the Armfield console shown in Figure 2. For experiment A the peristaltic feed pump labeled 3 will be calibrated. For Experiment B valves 2 and 3, labeled 7 and 8 respectably in Figure 1, will be used for the collection of samples to analyze the column efficiency. For Experiment C the location of the feed will be varied from the middle of the column (pictured) to the top.

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Commented [AZ3]: A steady-state binary distillation column.
What about changing feed position?

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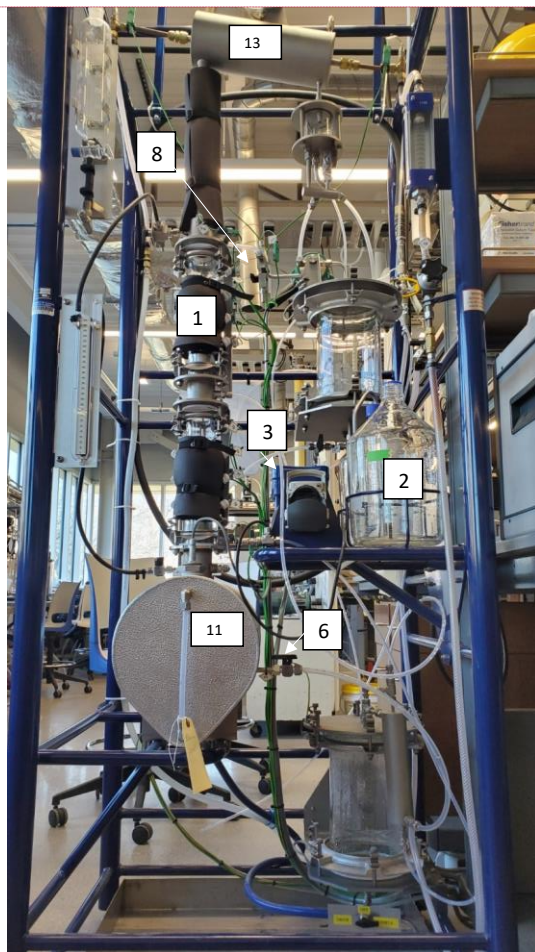


Figure 1 Distillation Column Apparatus

Commented [PTH5]: also, missing the cooling water for the condenser, the reflux, bottom product, etc.

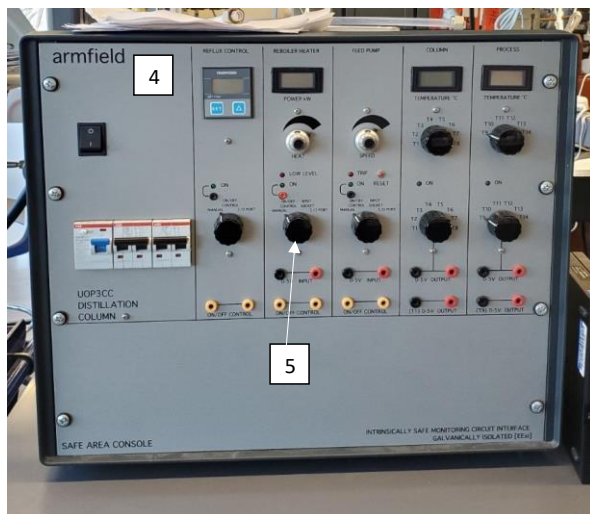


Figure 2 Armfield Console

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Table 1: The important components used in the experiment

#	The important components	Description
1	Distillation Column	Column where the components will separate
2	Feed Tank	
3	Feed pump	An electromechanical device used to recirculate the fluids through the piping structure, it designs to give 1ml/min per revelation
4	Armfield	Device used to control and collect temperature, reflux, and pump speed
5	Reflux switch	
6	V1	Used to allow the bottom product to exit via the heat changer and transfer to bottom product receiver
7	V2	Used to collect sample form the bottom
8	V3	Used to collect sample from the overheads
9	V5	
10	V10	Reflux valve
11	Reboiler	Used to provide heat to the bottom of the distillation column
12	Vacuum Pump	Used to keep the process in vacuumed pressure
13	condenser	Used to condensate the ethanol
14	receiver vessel	Used to collect ethanol mixture

Commented [PTH7]: there are 14 components listed here but you didn't label them all in the figure above. for example, missing 7, 9, 10, and 12 labels.

Commented [PTH8]: what's the description for this one?

2.2 Materials and Supplies:

Table 2: The important materials and supplies used in the experiment

#	The Material and Supplies	Description
1	Ethanol	Chemical compound had formula of (C ₂ H ₅ OH) used to make the binary mixture of water and ethanol
2	Water	Used to make the binary mixture of water and ethanol
3	graduated cylinder	Plastic cylinder used to measure volume of
4	stopwatch	Used to calibrate the pump and collect the samples
5	funnel/blue hose	Used to easily and safely load the mixture of water and ethanol into the reboiler.
6	Bottles	Contain the feed mixture.
7	blue cap	Used to
8	the black tubing	Used to connect the feed bottle to the feed pump
9	Fitting the flexible tubing	Inlet mixture feed
10	Cooling water	Used in the condenser

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3. Experimental Plan:

Preparing the solutions:

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- **Reboiler solution:** A solution of 10 L; 10 wt% ethanol and 90 wt% water.
- **Feed solution:** A solution of 3 L of 30 wt% ethanol solution, 70 wt% water

Experiment A: Feed Pump Calibration

Normally, the peristaltic feed pump is only design to preform $\sim 1\text{ml/ min}$ per revelation of the drive shaft, the pump is calibrated to acquire the accurate flowrate to the distillation column.

1. Disconnect the feed tubing connected to the distillation column
2. Make sure that the feed tank with pump section pipe has enough water for the calibration.
3. In order to determine the flowrate at the ten settings, use a graduated cylinder and stopwatch
4. Obtain the graph which can be found near the control console.

Experiment B: Steady State Distillation of an Ethanol-Water Mixture

Equipment Set Up

1. Switch off the reflux ration timer on the console to make sure that the equipment operates at total reflux
2. Make sure to get familiar with the location of each component in the unit by reading the Armfield instruction manual
3. Close all the valves located in the equipment.
4. On the reflux pipe, open valve (V10)
5. Using the funnel/blue hose located in the fume hood, load the 10 L of the mixture into the filler cap on the top of the reboiler.
6. Make sure that the filler cap located on the top of the reboiler is replaced after the filling is completed.
7. Using one of the feed bottles, fill it with 3 L of the 30 wt% mixture of ethanol and water
8. Now, make sure that the **blue cap** is connected to the feed bottle
9. Make sure that the **black tubing** is connected from the feed bottle to the feed pump
10. Make sure that the **clear feed tubing** is connected from the outlet of the pump to the column between Try 4 (T4) and tray 5 (T5)

The following instructions explain how to connect/ replace Feed Pump using flexible tubing

- Insert the black tubing into the slot located between the pump head and the rotor.
- Adjust/Move the Clamp forwards and downwards in order to close the pump head.
- After that, the Clamp will Click Shut
- In order to prevent permanent deformation of the tube, make sure to release the Clamp if the equipment is not used.

Procedure:

1. Turn on the power of the control panel
2. Adjust the temperature selector switch to **T9** and the **temperature inside the reboiler**
3. At flow rate of FI1 of ~ 3 L/min, make sure that valve (V5) is open in order to allow the cooling water to the condenser
4. **If needed,**
5. On the control panel, turn the power controller for the reboiler heating element fully anti-clockwise
6. Make sure to switch the switch responsible for turning on the power to the heating element to be read (**Power on**) on the control panel
7. Make sure to check that the red lap illuminates which is another way to check that heating element is on
8. Now, turn the controller clockwise until on the digital wattmeter, a reading of ~ 1.5 kW is obtained
9. Make sure to check the temperature readout meter to observe that the content of reboiler begins to heat up
10. If you want to observe the increase in temperature, switch the temperature selector to T8, T7, T6, T5, T4, T3, T2, T1
11. Turn the controller of the reboiler power to be 0.75 kW
12. Now, make sure to start the experiment with total reflux
13. before switching on the reflux switch, make sure to set the reflux ratio to 5:1 (meaning 5 second back to the column and 1 second to top product receiver)
14. Now, on the control panel, switch on the reflux valve
15. Make sure to hear the sound of click which indicates that reflux valve is working
16. Wait $\sim 15 - 30$ min before the column is stabilized at total reflux
17. After that, start of the bottom product flow stream, flow of the feed and reflux can begin
18. Open valve (V1) to allow the bottom product to transfer into the bottom product receiver through the heat exchanger.
19. **It is preferred,** to set the feed flow to be $\sim 30 - 35$ ml/min (obtained from the feed pump calibration curve)
20. Approximately after feed ~ 500 ml, **at the same time,** take a sample of overhead using valve (V3) and sample of the bottom product through valve (V2)
21. Record the temperature in the column (T1 – T9)
22. Record the refractive index for the collected samples

23. Make sure to repeat steps (20 – 22) every 5 minutes to collect more than two samples
24. Turn off the feed pump, the reflux switch, the reboiler power after complete collecting the sample
25. Make sure to close valve (V1) at experiment B is done

Experiment C: Varying The Feed Position

Changing Feed Position

1. Make sure to disconnect the feed tube connecting the column at the midpoint between T4 and T5
2. Make sure to disconnect the plug from the feed position (above the top plate (T1))
3. Now, reconnect the plug to the column at midpoint between (T4 and T5)
4. Then connect the feed tube to the column at the feed position (above top plate (T1))

Procedure

1. Turn on the reboiler power to 0.75kW (make sure to have stable total reflux before starting.
2. bottom product flow, the flow of feed and the reflux should be started at the same time (5min to heat up)
3. open valve (V1) to let the product exit to bottom product receiver (The feed to the column will now be admitted at the top of the column)
4. Set the feed to be 30-35 milliliters/min
5. Take a sample from valve (V3) and (V2) after 500 milliliters been fed, repeat that to collect 3 samples from top and bottom. (5 minutes apart, record the temperatures from T1 and T9)

- **Independent Variables:** cool water flowrate, times solution composition
- **Dependent Variables:** gas and vapor composition, temperature stages, reflux rate
- **Calibration Methods:** refractometers by using refractory index solution
- **Satisfaction Methods:** No need for this step because only one trial needs to be done
- **Goodness of fit:** the comparison will be with McCabe Thiele Diagram to observe the good fit with the binary batch distillation.

Project Safety Evaluation

Project Title: Continuous Distillation

Date: 02/18/2020

-	Yes/No	
Potential electrical hazards?	Yes	High voltage from heating source element
Potential mechanical hazards?	Yes	Falling hazards from heights more than 6 feet
Hazardous/toxic chemicals involved?	Yes	Toxic chemicals if inhaled large amounts
Incompatible chemicals involved?	No	-NA-
Extra eye protection required?	Yes	Goggles
Gloves required?	Yes	Nitrile gloves
MSDS reviewed by all team members?	Yes	UN1230
Temperature (°C)	100	Reboiler, column
Pressure (psig)	14.7	column
Safety Controls	yes/no	If yes, Provide description
	yes	Switch power off

Commented [PTH13]: 7.5/10

Commented [PTH14]: and it is located near water source

Commented [PTH15]: will you be anywhere 6ft off the ground?

Commented [PTH16]: where are the potential pressure and temperature hazards?

Commented [PTH17]: what is this? ethanol? water?

Commented [PTH18]: and fume hood for preparing ethanol solution

I have read relevant background material for the Unit Operations Laboratory entitled: “Continuous Distillation” and understand the hazards associated with conducting this experiment. I have planned out my experimental work in accordance to standards and acceptable safety practices and will conduct all of my experimental work in a careful and safe manner. I will also be aware of my surroundings, my group members, and other lab students, and will look out for their safety as well.

Signature of Team member and Date

Ahmad Abusubaih 02/18/2020

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